

Chapter 5: Proposed Method

RS-DRL-Based Self-Adaptive System

Thesis Documentation

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1 Proposed Method: RS-DRL-Based Self-Adaptive System

2 Proposed Method: RS-DRL-Based Self-Adaptive System

Figure ?? introduces a diagram map that organizes the process models presented in this chapter. The central integration diagram provides a high-level view of system behavior, while detailed activity diagrams refine individual subprocesses.

Figure ?? presents the high-level integration process overview, showing the nine numbered stages that characterize the system's operation.

Figure ?? presents the integrated process view, which refines the architectural view introduced in the reference model by making control flow, data flow, and feedback loops explicit.

3 Key Components and Their Responsibilities

3.1 Monitoring and State Construction

Figure ?? details the monitoring and state construction process, which represents the entry point of uncertainty into the system.

3.2 RS-DRL Module: Training and Learning Components

Figure ?? presents a detailed view of the RS-DRL training process, which constitutes the core learning mechanism of the proposed approach.

3.3 Adaptation Option Prediction

Figure ?? illustrates the runtime adaptation option prediction process. Unlike the training process, this path is strictly non-learning and is invoked only when adaptation is required.

3.4 MAPE-K Adaptation Control Loop

Figure ?? refines the adaptation control logic using the MAPE-K framework, emphasizing that adaptation is event-driven rather than continuous.

3.5 Adaptation Option Verification

Figure ?? presents the adaptation option verification process, which provides semantic grounding and trustworthiness for learning-based decisions.

4 Novel Reward Shaping Mechanism in RS-DRL

4.1 Reward Shaping Function

Figure ?? isolates the proposed reward shaping mechanism applied during RS-DRL training.

Diagram Map of RS-DRL Process Models

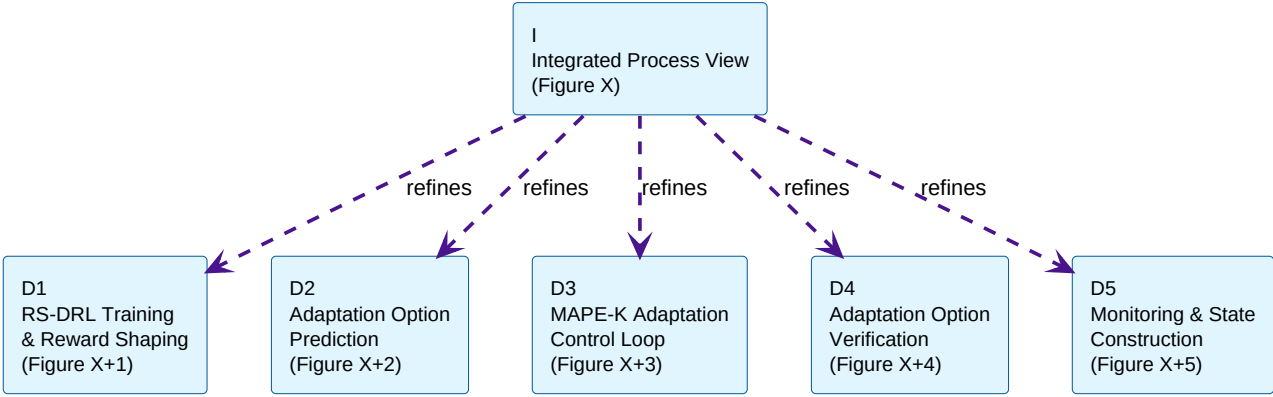


Figure 1: Diagram map of the RS-DRL-based self-adaptive system. The central integration diagram provides a high-level process view, while Diagrams D1–D5 refine the learning, decision-making, verification, control, and monitoring subprocesses presented in this chapter.

4.2 RS-DRL Training Algorithm

4.3 Design Rationale and Comparison

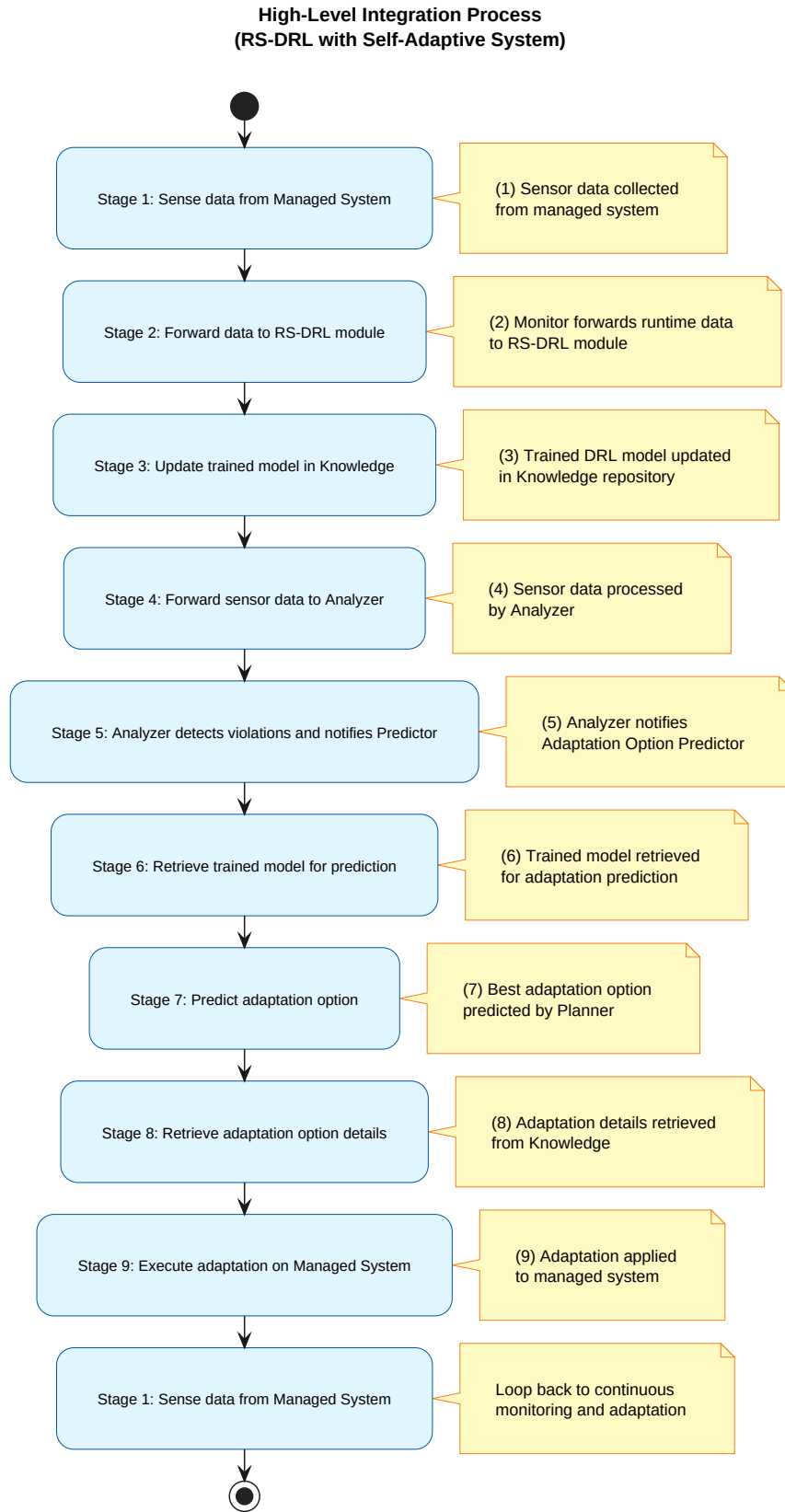


Figure 2: High-level integration process overview showing the nine numbered stages of the RS-DRL-based self-adaptive system. The sequential flow illustrates how data sensing, model updates, analysis, planning, and execution are integrated in a continuous adaptation loop.

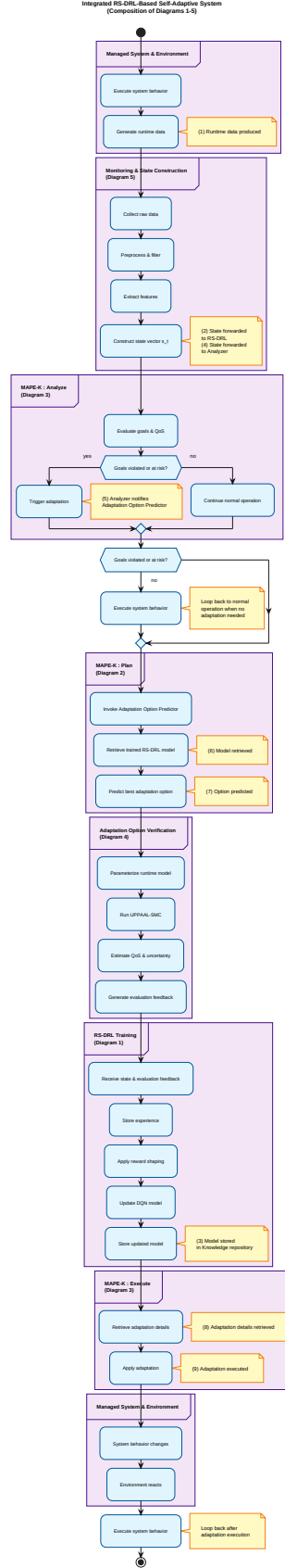


Figure 3: Integrated process view of the RS-DRL-based self-adaptive system. The figure illustrates the closed-loop interaction between monitoring, learning, verification, planning, and execution, and refines the original architectural view presented in Figure 4 of the reference model.

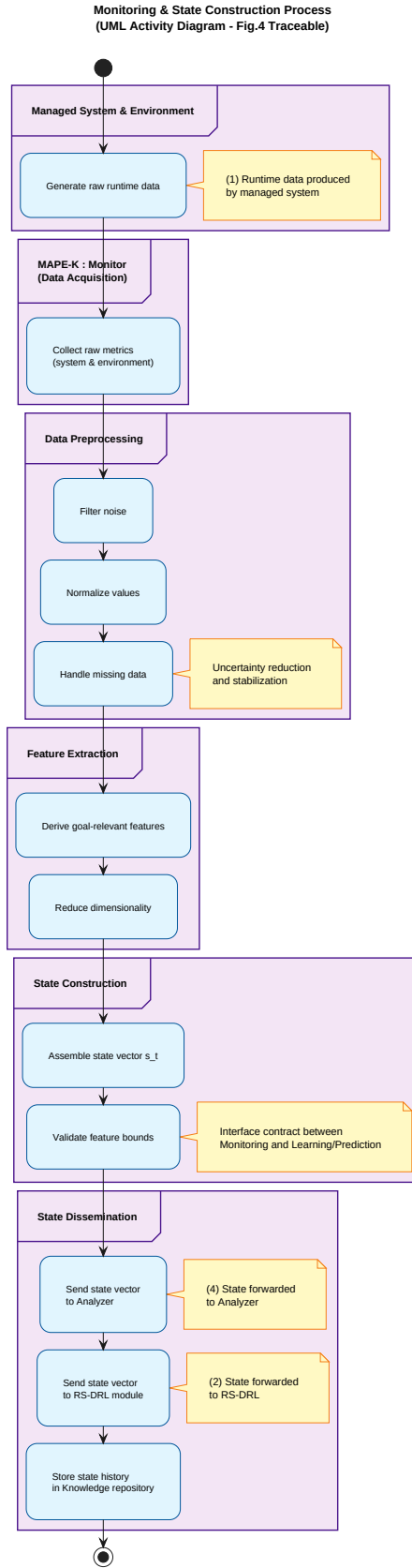


Figure 4: Monitoring and state construction process. Raw runtime data from the managed system and environment is filtered, abstracted, and transformed into a state vector suitable for both RS-DRL learning and runtime adaptation analysis.

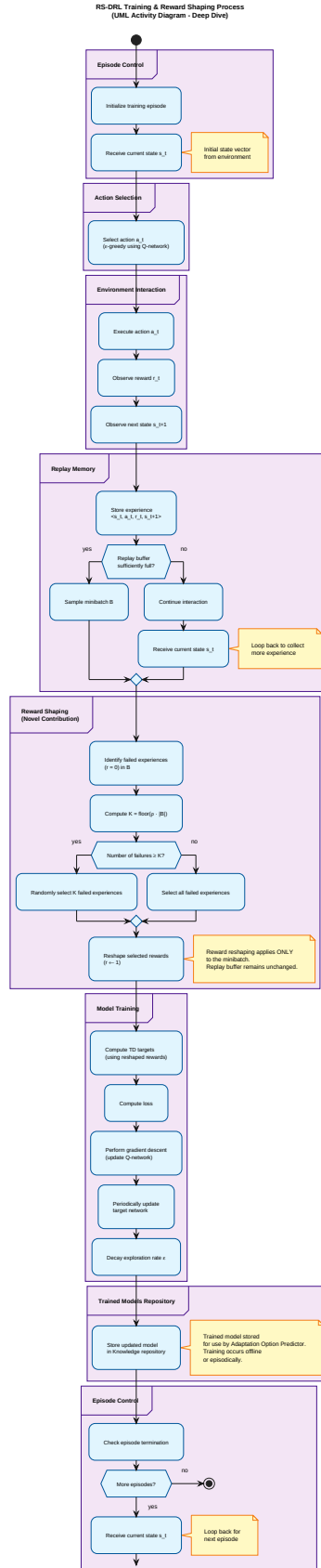


Figure 5: RS-DRL training and reward shaping process. The figure details the episodic interaction with the environment, experience replay, and the novel reward reshaping mechanism applied during minibatch learning.

Adaptation Option Prediction Process
(Runtime Decision Path - Fig.4 Traceable)

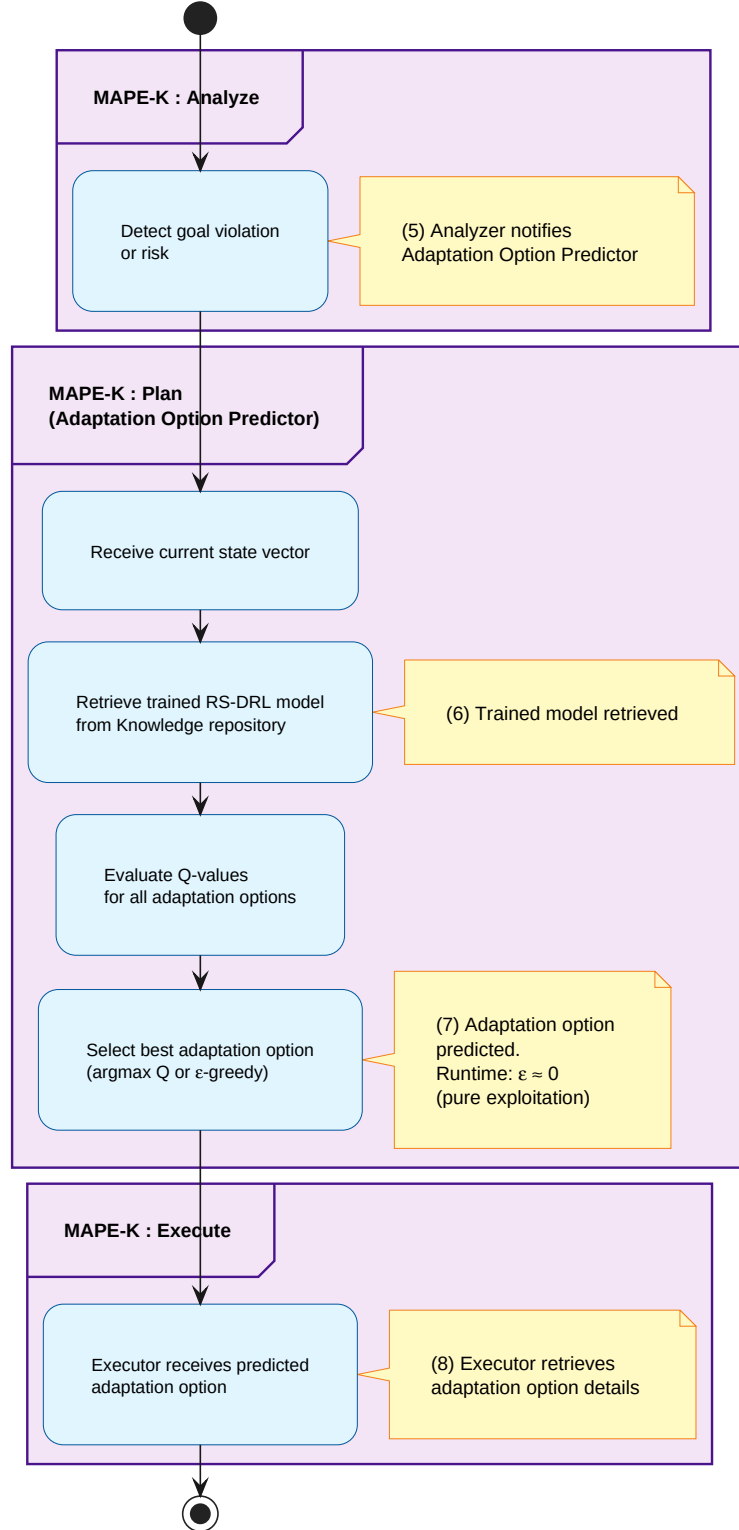


Figure 6: Adaptation option prediction process at runtime. Upon detection of goal violations, the trained RS-DRL model is used to predict the most suitable adaptation option without further learning.

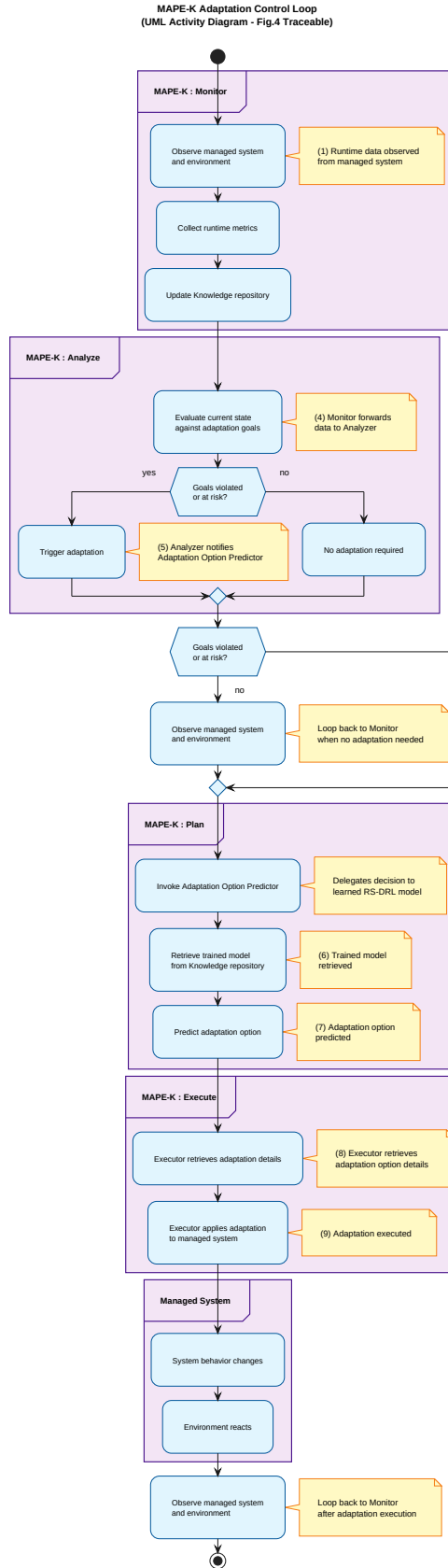


Figure 7: MAPE-K adaptation control loop. The figure illustrates how monitoring and analysis trigger adaptation, how planning delegates decision making to the RS-DRL predictor, and how execution enacts the selected adaptation.

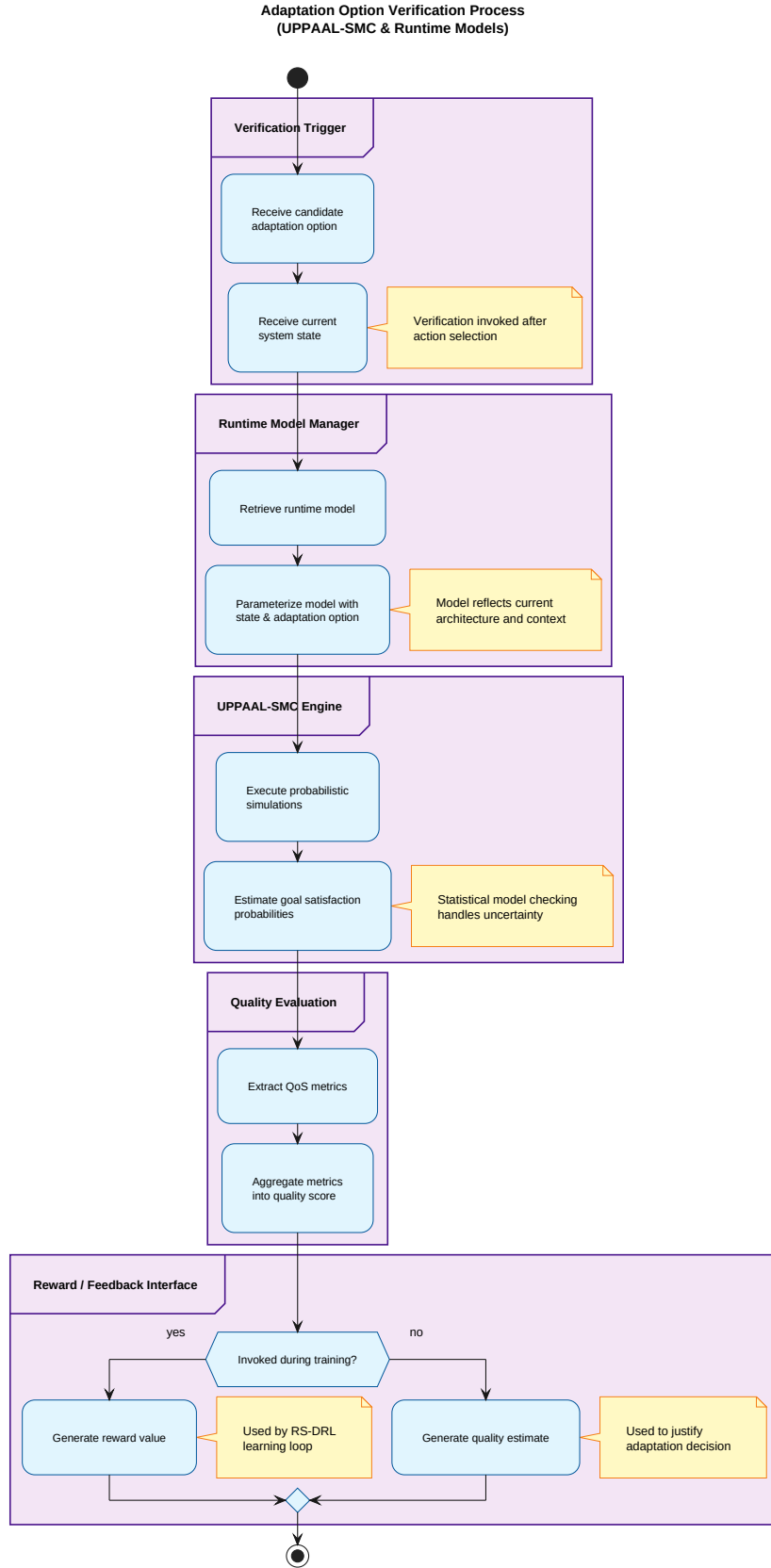


Figure 8: Adaptation option verification using runtime models and statistical model checking. The verification process evaluates the expected quality of adaptation options under uncertainty and provides feedback for learning and decision justification.

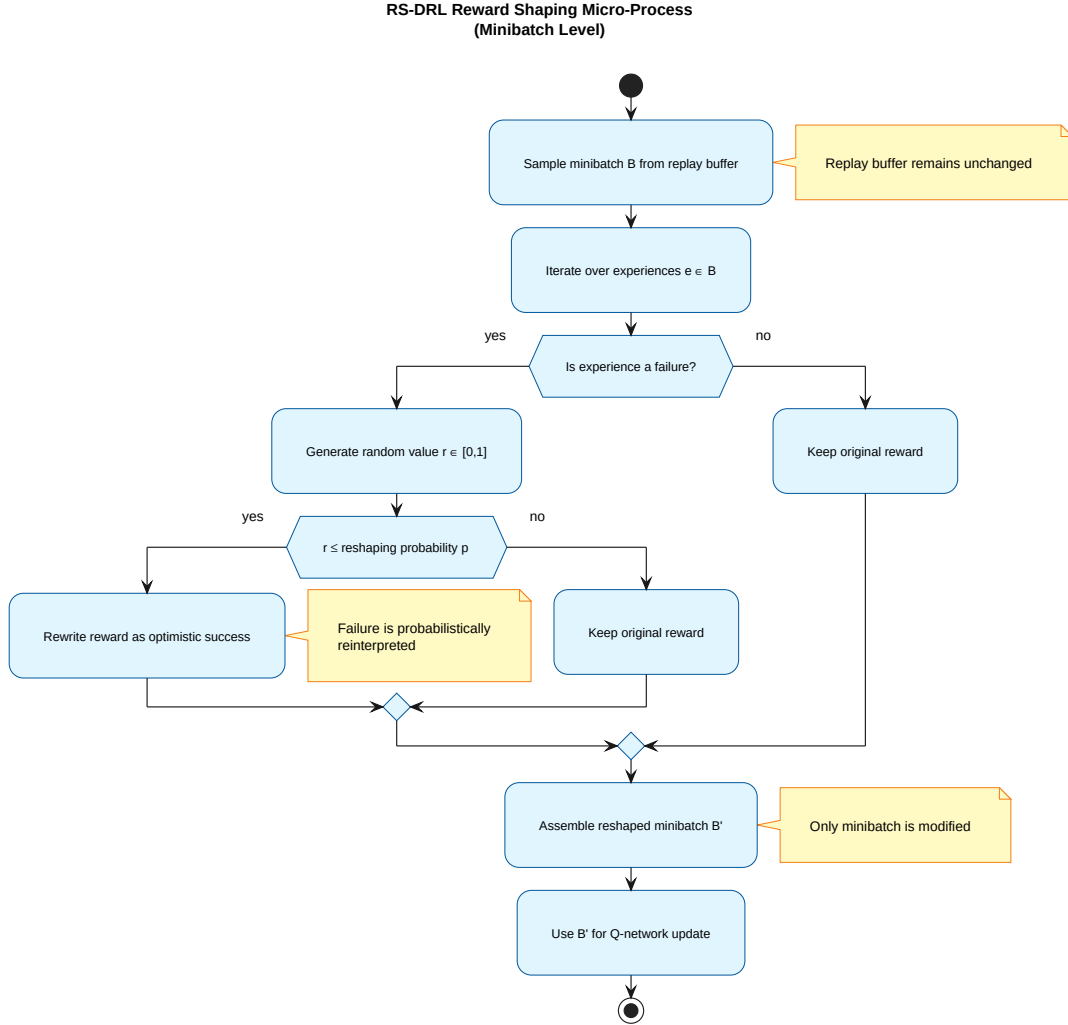


Figure 9: Reward shaping micro-process applied during RS-DRL training. Failed experiences sampled in a minibatch are probabilistically reinterpreted as optimistic successes according to a reshaping probability p . Reward reshaping is applied exclusively to the sampled minibatch and does not modify the replay buffer, preserving experience diversity while accelerating learning convergence.